
Herd Immunity and Learning Externalities in Markets of Adaptive Models

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Abstract

1 A learning agent can pass every single-agent stability test and still help destabilize
2 the market it runs in. When agents share an environment that reacts to their
3 decisions, each one’s retraining reshapes the data the others learn from next, and
4 no agent internalizes that effect. We call it a *learning externality*. Its strength is the
5 market’s systemic modulus m_N , the factor by which a common perturbation grows
6 per retraining round. What sets m_N is not the number of firms but the *effective*
7 *number of independent learners*, the leading eigenvalue of the correlation matrix
8 of firms’ response Jacobians. Foundation-model concentration therefore enters
9 the stability condition as a term rather than as a concern: fifty firms fine-tuning
10 one vendor’s model are dynamically close to one large learner. Three closed-form
11 levers follow from a reduction lemma we derive in the main text. Taking fewer
12 gradient steps between deployments buys stability, but only up to a critical crowding
13 level past which no cadence is safe. Feedback-aware correction reproduces the
14 imperfect-vaccine coverage law, stabilizing the market once the corrected fraction
15 exceeds $(1 - 1/m_N)/e$ at efficacy e , a form exact in the fully shared limit and
16 accurate away from it. A Pigouvian wedge prices the externality, and every market
17 of two or more firms over-adapts. The last two levers are substitutes along a frontier
18 we compute. In a shared-pool order-flow market the mechanism reproduces the
19 $1.74\times$ and $3.16\times$ amplification of a recent preprint, bit for bit.

20 1 Introduction

21 A learning agent can pass every stability test we know how to run and still help destabilize the market
22 it operates in.

23 The test we know how to run is a single-agent test. A model is evaluated, stress-tested and certified
24 alone, against a distribution treated as given. When the deployed model reshapes that distribution, the
25 performative prediction literature supplies the correction: repeated retraining converges if and only if
26 a modulus $m = \varepsilon\beta/\gamma$ stays below one, where ε measures how strongly the environment responds
27 and γ is the learner’s own objective curvature (Perdomo et al., 2020). Certify $m < 1$ and the loop is
28 safe.

29 It is not, once the agent has company. Consider dealers whose quotes reshape a common pool
30 of informed flow, recommenders competing for one attention pool, or lenders pricing against one
31 population of borrowers. Each agent’s retraining changes the data every *other* agent will learn from
32 next, and no agent internalizes that effect. We call this a **learning externality**: every participant
33 behaves rationally, passes its own test, and helps amplify a loop none of them chose. The intuition is
34 not new. Banks that each diversify optimally hold similar portfolios and fail together (Beale et al.,
35 2011; Wagner, 2010), and funds running similar strategies unwound together in August 2007 with

the crowding invisible from any single risk report (Khandani and Lo, 2007). What is new is that the agents now *learn*, which changes both what determines the danger and what can be done about it.

What determines it is not how many firms there are. Let firm i have a response Jacobian E_i describing how its own deployment reshapes the flow it faces, and let R be the correlation matrix of the vectorized E_i . We show the market’s stability is governed by $m_N = N_{\text{eff}} m_1$ with $N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1)$, where κ is the spillover between firms and $\lambda_{\max}(R)$ is the **effective number of independent learners**. With s the fraction of each response attributable to a common model, vendor or pretraining corpus, this reads $m_N = m_1(1 + \kappa s(N - 1))$, so **the effective number of learners is the number of independent models, not the number of firms**. The quantity a competition regulator measures is the wrong one: fifty equal-share firms have a minimal Herfindahl index and look perfectly competitive, and if all fifty share one vendor they are dynamically a monoculture (Hirschman, 1964).

What can be done about it takes three forms, and two are substitutes. Because every result is stated in m_N , each inherits the supply chain for free. *Slow down*: K -step retraining faces a window $K < K_{\max}(m_N)$ that narrows as crowding rises and shuts past a critical crowding level, beyond which no cadence is safe. *Correct*: feedback-aware updates (Izzo et al., 2021) run inside the market produce exactly the epidemiological herd-immunity law with the systemic modulus as the reproduction number. *Diversify*: that threshold rises in s , so a market reaches stability along either axis, and the iso-stability frontier between the two is this paper’s most directly usable object. Pricing the externality completes the policy triple, and the decentralized equilibrium over-adapts for every $N \geq 2$.

Contributions. (1) The effective number of independent learners as a market’s reproduction number, derived from a reduction lemma, which turns foundation-model concentration into a term in a stability condition and invalidates mean-similarity diversity indices twice over. (2) The crowding-cadence frontier and its critical crowding level, composing lazy deployment with multi-agent amplification. (3) A mixed-market herd-immunity theorem, exact at finite correction strength, whose clean form is the imperfect-vaccine coverage law and which carries a critical efficacy. (4) The substitution frontier between diversity and corrected learning, which exists only once (1) and (3) are expressed in one quantity. (5) The Pigouvian wedge and the over-adaptation corollary.

2 Related work

Monoculture, and why it is static. Kleinberg and Raghavan (2021) show that when many firms adopt the same algorithm welfare can fall even when the shared algorithm is the better one, because correlated selection destroys the option value of independent evaluation; Bommasani et al. (2022) formalize systemic exclusion when many deployments inherit one model. Peng and Garg (2024) is the nearest neighbour and the first to carry market effects, a two-sided matching model in which many decision-makers share one evaluation algorithm; Jagadeesan et al. (2023) make the supply-side move, characterizing what producers supply when they compete through a shared recommender. All four are equilibrium statements, read at the market’s resting point. None expresses a stability boundary or a rate, and none carries an object for *which direction* a firm perturbs the environment, so a market can be statically fine and dynamically fragile at the same shared-model fraction.

Multiplayer performative prediction, and the containment relation. Narang et al. (2023) is the nearest dynamics: multiple learners in a shared decision-dependent environment, with equilibrium concepts and convergence conditions for repeated retraining. Their condition bounds scalars, the Lipschitz sensitivity of the joint distribution map against objective curvature, uniformly over the joint action. Ours bounds a spectral radius in which those scalars appear only inside m_1 . The relation is strict refinement rather than corollary, and Proposition 3 gives the witness: two markets agreeing on every scalar of that kind, differing in stability, and so indistinguishable to any criterion built from those scalars alone. The substitution $\mathbf{1}\mathbf{1}^\top \rightarrow R$ adds an object rather than tightening a constant, and that object carries monoculture, provenance, the supply chain and every policy statement below. Two further multi-agent treatments border this one. Piliouras and Yu (2023) follow many predictors of one common outcome from global stability through instability into formal chaos as learning rates and agent counts grow, which is the monoculture corner $R = \mathbf{1}\mathbf{1}^\top$ carried past its linearization

rather than one configuration among many. Li et al. (2022) schedule deployments for decentralized agents coupled by a consensus constraint, so their coupling is chosen cooperatively and no externality arises. The single-learner lazy-deployment slope of Mendler-Dünner et al. (2020) is what Theorem 4 composes with amplification no firm chooses, a composition not previously made; the corrected update of Izzo et al. (2021) is what Theorem 5 puts into the game, having only ever been run single-learner.

Systemic risk, epidemiology, and the statistic. Beale et al. (2011) asked this of banks, but their agents do not learn, so the homogeneity is portfolio overlap rather than alignment of feedback directions and there is no second instrument to trade diversity against. Diekmann et al. (1990) converts the epidemiological parallel from analogy into identity: R_0 in a heterogeneous population is *defined* as the spectral radius of a next-generation operator, and m_N is the spectral radius of a linearized joint retraining operator, so the two thresholds are the same statement and the vaccination law transfers with its refinements. The largest eigenvalue of a correlation matrix as an effective count is meanwhile standard in random-matrix finance (Laloux et al., 1999; Plerou et al., 2002), signal processing (Roy and Vetterli, 2007) and ecology (Hill, 1973); the novelty is the matrix it is computed from, response Jacobians rather than returns, and the condition it enters.

3 Setup: private against systemic stability

The base result. Consider N symmetric dealers quoting the same instruments and sharing one pool of informed order flow, coupled by a spillover $\kappa \in [0, 1]$: one dealer tightening its quotes raises the toxic flow routed to every dealer, in proportion κ . Each retrains periodically on the flow it has observed, and in isolation a firm is stable exactly when its performative modulus satisfies $m_1 < 1$. The joint retraining map has Jacobian $J = -m_1[(1 - \kappa)I + \kappa \mathbf{1}\mathbf{1}^\top]$, whose dominant common mode sits at $-m_1 N_{\text{eff}}$ with $N_{\text{eff}} = 1 + \kappa(N - 1)$, so the market crosses into instability a factor N_{eff} earlier than any individual dealer’s own loop would. Measured in a shared-pool simulator with real spreads, inventory and liquidity feedback, amplification is $1.74\times$ at $N = 2$ and $3.16\times$ at $N = 3$ against a linear prediction of 2 and 3 (Nagarajan and Ashok, 2026). This is inherited scaffolding; everything below replaces one object in it.

Two questions, and they are not the same question. Is each agent stable alone, $m_1 < 1$? And is the collection stable when its members interact, $m_N < 1$? A shared environment separates them, because the loop closes through that environment and nothing in dealer A’s validation set contains dealer B. **Agent count is not a descriptive characteristic of a market; it enters the stability condition.** Read as economics, this is a *technological* rather than pecuniary externality (Buchanan and Stubblebine, 1962): a price movement is a transfer, but reshaping the distribution a competitor learns from changes everyone’s forecasting technology, which is a real efficiency loss. We call m_N the **feedback reproduction number**: the factor by which a common perturbation is amplified per retraining round.

Objects and standing assumptions. Firm i has a response Jacobian $E_i \in \mathbb{R}^{d \times d}$. The *alignment matrix* $R = (r_{ij})$ collects $r_{ij} = \langle \text{vec } E_i, \text{vec } E_j \rangle / (\|E_i\|_F \|E_j\|_F)$, a correlation matrix of feedback directions rather than of returns; it is positive semidefinite with unit diagonal.

Assumption 1 (Standing). (A1) All spectral statements linearize the joint retraining map around the joint equilibrium, with the simulator as the nonlinear check. (A2) Theorem 2 is exact for equal moduli $m_i = m_1$; unequal moduli give a two-sided bound. (A3) Firms retrain on a common clock, taking a common number K of gradient steps between deployments. K is the cadence throughout: a larger K means a larger move per round, not more frequent retraining. (A4) Firms couple through one pool with a scalar spillover κ . No sign condition on R is assumed; the one place one is needed is stated where it is used, in Section 7.

The reduction. The substitution of R for $\mathbf{1}\mathbf{1}^\top$ is the whole paper, so it is derived rather than posited.

Lemma 1 (Reduction). Assume (H1) firm i ’s deviation from equilibrium is a scalar x_i along its unit response direction $\hat{E}_i = E_i / \|E_i\|_F$, so the state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$; (H2) $\|E_i\|_F = \varepsilon$ for every i , which is (A2) in the new object; (H3) firm i retrains against the component of the pool

distortion along $\hat{E}_i$, feeling its own contribution in full and each competitor's with weight κ ; and (H4) firm i best-responds to the felt distortion under a γ -strongly-convex objective of sensitivity β , so $x_i^+ = -(\beta/\gamma) \text{felt}_i$. Then the joint retraining map is linear with Jacobian

$$J = -m_1[(1 - \kappa)I + \kappa R], \quad m_1 = \varepsilon\beta/\gamma. \quad (1)$$

Proof sketch. The pool carries the aggregate distortion $Z = \sum_j x_j E_j$, so by (H3) firm i feels $\text{felt}_i = \langle \hat{E}_i, x_i E_i \rangle + \kappa \sum_{j \neq i} \langle \hat{E}_i, x_j E_j \rangle$. By (H2), $\langle \hat{E}_i, E_j \rangle = \varepsilon r_{ij}$, and since $r_{ii} = 1$ the own term folds into the sum, leaving $\text{felt}_i = \varepsilon[(1 - \kappa)x_i + \kappa(Rx)_i]$. Applying (H4) coordinatewise gives (1), with m_1 the single-firm modulus of Perdomo et al. (2020). $\square$

Alignment enters through (H3), and in the market this model comes from the hypothesis is literal rather than convenient. A dealer observes order flow only where it quotes. Flow that a competitor's repricing pushes into instruments the dealer does not trade never reaches its training set, so that competitor's perturbation is invisible to it rather than merely small, however large it is. Own-channel sensing is what makes R appear at all: two firms perturbing the environment in orthogonal directions do not read each other. (H1) is the hypothesis that limits scope; without it J carries a Kronecker structure on $\mathbb{R}^{Nd}$ whose radius is not in general $N_{\text{eff}}m_1$, the block-secular problem deferred in Section 9.

4 Result 1: the effective number of independent learners

Counting firms is wrong, and the correction is where monoculture and supply-chain content enter. $\lambda_{\max}(R)$ runs from 1, all responses orthogonal, to N , monoculture.

Theorem 2 (Alignment sets the reproduction number). *Under Assumption 1 with equal moduli and heterogeneous response directions, the joint Jacobian is (1) and the market's spectral radius is*

$$m_N = N_{\text{eff}} m_1, \quad N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1), \quad (2)$$

so the market is stable if and only if $m_N < 1$.

Setting $R = \mathbf{1}\mathbf{1}^\top$ recovers the symmetric multi-dealer law of Nagarajan and Ashok (2026) exactly, so the earlier result is not generalized away but located: it is the monoculture corner, where every firm's feedback points the same way. Four anchors follow, each checked independently and derived in Appendices B and C. Monoculture gives $N_{\text{eff}} = 1 + \kappa(N - 1)$, the inherited law. Orthogonal responses, $R = I$, give $N_{\text{eff}} = 1$: a hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm, and the simplex $r_{ij} = -1/(N-1)$ gives $1 + \kappa/(N-1)$. A supply chain of share s gives $N_{\text{eff}} = 1 + \kappa s(N - 1)$, the number of independent *models* rather than of firms. $N_{\text{eff}} \geq 1$ always, so interaction never stabilizes a market below what its members achieve alone.

Proposition 3 (Strict refinement, with a witness). *Fix $N \geq 2$, $\kappa \in (0, 1]$ and m_1 with $1/(1 + \kappa(N - 1)) < m_1 < 1$. Compare the market at $R = I$ with the market at $R = \mathbf{1}\mathbf{1}^\top$. The two agree on every scalar a uniform Lipschitz-and-curvature criterion can read: the same response magnitude ε by (H2), the same objective curvature γ and sensitivity β by (H4), hence the same m_1 and the same spillover κ . Their radii are $m_N = m_1$ and $m_N = m_1(1 + \kappa(N - 1))$, which the choice of m_1 places on opposite sides of one. Any criterion measurable from those scalars alone therefore returns the same verdict for both, and is wrong about one of them.*

At $N = 20$, $\kappa = 0.8$ and $m_1 = 0.15$ the orthogonal market runs $m_N = 0.15$ and the monoculture $m_N = 2.43$: one converges, the other diverges, and no Lipschitz constant in either model distinguishes them. The object doing the separating is R , which no scalar criterion carries, so Theorem 2 refines a uniform condition rather than following from one.

The supply chain. Decompose $E_i = \sqrt{s} E_{\text{shared}} + \sqrt{1 - s} \Xi_i$, where $s \in [0, 1]$ is the fraction attributable to a shared foundation model, vendor or pretraining corpus. Error correlation across more than 350 hosted models is substantial and concentrated within shared providers (Kim et al., 2025), so the premise of this decomposition is observed on deployed systems rather than assumed. Alignments concentrate at $r_{ij} \rightarrow s$ with $O(1/d)$ fluctuation and an explicit failure probability, so $N_{\text{eff}} \rightarrow 1 + \kappa s(N - 1)$ and every result below inherits s for free: fifty dealers fine-tuning one vendor's model are dynamically $\sim 1 + 49\kappa$ learners at $s \approx 1$ and barely more than one at $s \approx 0$.

186 **Why the naive diversity index fails, twice.** Mean alignment understates *clustered* alignment.
187 Three tightly aligned firms among ten otherwise-orthogonal ones give $N_{\text{eff}} = 2.60$ at $\kappa = 0.8$,
188 against 1.48 for the uniform configuration with the identical mean: the market runs $m_N = 1.30$ at
189 $m_1 = 0.5$ and is unstable, while the index reports 0.74 and calls it safe with margin. The simplex is
190 sharper still, minimizing mean alignment while carrying $\lambda_{\text{max}} = N/(N-1) > 1$, so the mean fails
191 as a *ranking* and not only as a level. And the failure is signed: $N_{\text{eff}} \geq 1 + \kappa(N-1)\bar{r}$ for every R , so
192 the index never over-states risk (Appendix B).

193 5 Result 2: the crowding-cadence frontier

194 The first lever is quantity regulation, and it is the one a firm can pull alone: move less per round. A
195 learner taking only K gradient steps per deployment realizes outer-map slope $\mu(K) = -m + c^K(1 +$
196 $m)$, where $c = 1 - \eta\gamma$ is the inner per-step contraction (Mendler-Dünner et al., 2020). Because c
197 is built from the learner’s own objective curvature and step size, it does not depend on how many
198 competitors it has, and that independence is what makes the composition below work. We verify it by
199 measuring the realized contraction inside the joint market.

200 **Theorem 4** (Systemic cadence window). *Under Assumption 1, in the N -firm market with synchronous*
201 *K -step retraining the joint deployment map is $c^K I + (1 - c^K)J$, so the binding mode has slope*
202 *$\mu_N(K) = -m_N + c^K(1 + m_N)$ and the market is stable if and only if*

$$K < K_{\text{max}} = \frac{\ln((m_N - 1)/(m_N + 1))}{\ln c}, \quad (3)$$

203 *with K_{max} infinite when $m_N \leq 1$ and strictly decreasing in m_N .*

204 The slope on the mode carrying $\lambda_i(R)$ is strictly decreasing in λ_i , which puts every slope at or below
205 $c^K < 1$, so the upper side of the constraint never binds at any cadence, and places the binding mode
206 at λ_{max} whatever the sign pattern of R .

207 **Critical crowding.** The same inequality reads $m_N < (1 + c^K)/(1 - c^K)$, and since K is a
208 positive integer the most permissive case is $K = 1$. A stabilizing cadence exists at all if and only
209 if $m_N < (1 + c)/(1 - c)$, a factor of 9 at $c = 0.8$. Past that the market diverges at every cadence,
210 however small the step budget. Moving less per round is the only unilateral defence, and past the
211 critical level it runs out.

212 **What the supply chain does to the window.** Substituting Result 1 makes K_{max} a function of
213 vendor concentration at fixed firm count: at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, the window
214 runs 20.68 steps at $s = 0.25$, 5.28 at $s = 0.5$ and 2.53 at $s = 1$. Raising vendor concentration cuts
215 every incumbent’s step budget eightfold while its competitors do nothing at all. The externality in its
216 sharpest form is not that **a competitor’s entry consumes your step budget**, but that **a competitor’s**
217 **choice of vendor does too**, without their entering.

218 **What the lever costs.** Cadence buys stability with staleness, and the exchange rate is the same
219 quantity on both sides: after a retraining round the deployed model’s gap to its own best response is
220 exactly c^K times what it was, so the c^K that widens the stability margin is the staleness.

221 6 Result 3: herd immunity, and diversity as its substitute

222 The second lever is a technology mandate, and unlike cadence it is not something a firm can pull
223 alone to any effect. A *corrected* firm runs the feedback-aware update (Izzo et al., 2021; Miller et al.,
224 2021), so its retraining is governed by the objective curvature γ_{PO} in place of the cobweb and its
225 modulus is scaled by $\theta = \gamma/\gamma_{\text{PO}} < 1$. What is new is running it inside the game: a mixed market of
226 N_b blind firms and $N - N_b$ corrected ones, corrected fraction $\rho = 1 - N_b/N$. Correction changes a
227 firm’s gain and not its response direction, so R is untouched.

228 **Theorem 5** (Mixed-market stability). *Under Assumption 1, with supply-chain alignment and two*
229 *correction levels, the joint Jacobian’s symmetric congruence is a diagonal matrix plus a positive*
230 *rank-one term, so its spectrum solves a secular equation. Two distinct moduli collapse it to a*



Figure 1: Left: the (N, s) phase diagram over 48 cells (Theorem 2). Right: the cadence window over 175 cells of the (N_{eff}, K) grid against predicted K_{max} , with s contours and $c = 0.8$ (Theorem 4); the realized window is $\lfloor K_{\text{max}} \rfloor$ and the plotted frontier continuous. Both are dry runs (Section 8).

231 *quadratic, and for $1 \leq N_b \leq N - 1$ the radius is its larger root,*

$$\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q}), \quad P = (1 - \tilde{\kappa})m_1(1 + \theta) + \tilde{\kappa}m_1(N_b + \theta N_c), \quad Q = \theta m_1^2(1 - \tilde{\kappa})N_{\text{eff}}, \quad (4)$$

232 *with $\tilde{\kappa} = \kappa s$ and $N_c = N - N_b$. An empty block needs the single-block form instead, the quadratic*
 233 *otherwise leaving a phantom root that can exceed the true radius.*

234 N_{eff} appearing inside Q is why every statement below carries s : the mixed market inherits Result 1's
 235 crowding through the product term, not by stipulation.

236 **The strong-correction limit, and it errs in the unsafe direction.** As $\gamma_{\text{PO}} \rightarrow \infty$ the constant
 237 term vanishes and the unstable cycle runs through blind firms only, giving $N_b < N_c(s) = 1 +$
 238 $(1/m_1 - 1)/(\kappa s)$. That criterion taken alone is **optimistic rather than conservative**. The radius is
 239 nondecreasing in θ by Perron-Frobenius on an entrywise-nonnegative matrix, so the limit under-states
 240 it: on random draws it calls 11.8% of configurations stable that are unstable at every finite correction
 241 strength. The exact root is therefore the criterion and the limit its corollary, quoted with its error
 242 direction. The same monotonicity carries the reassuring half: **correction never backfires**, so no
 243 configuration exists in which un-blinding a firm destabilizes the market.

244 **The collapse, and its honest generalization.** At $\kappa = s = 1$ the limit reads $\rho^* = 1 - 1/m_N$,
 245 **exactly the epidemiological herd-immunity threshold** $1 - 1/R_0$ with the systemic modulus as the
 246 reproduction number. A market of 10 firms at $m_N = 2.5$ needs 60% of them un-blinded *if correction*
 247 *is perfect*. It is not, and at that same corner the quadratic degenerates and the stability condition
 248 solves exactly for

$$\rho^*(e) = \frac{1 - 1/m_N}{e}, \quad e = 1 - \gamma/\gamma_{\text{PO}}, \quad (5)$$

249 with e the **correction efficacy**. That is the standard coverage requirement for an *imperfect vaccine*,
 250 of which the clean law is the perfect-efficacy corner. The clean form assumes homogeneous mixing
 251 (Fine et al., 2011), and the collapse here assumes the $\kappa = s = 1$ corner for precisely that reason.

252 **Corollary 6** (Critical efficacy). $\rho^*(e)$ *exceeds one, so no corrected fraction works at all, exactly*
 253 *when $\theta > 1/m_N$. Correction is a usable lever only if $\gamma_{\text{PO}}/\gamma > m_N$.*

254 At $m_N = 2.5$ the corrected update must deliver a $2.5\times$ curvature improvement or un-blinding cannot
 255 stabilize the market whatever fraction is treated. This parallels Section 5's critical crowding: each
 256 lever has a regime past which it stops working, and knowing both is what makes trading them off
 257 honest rather than an extrapolation.

258 **The synthesis.** ρ^* increases in s wherever it is positive, so a market reaches stability along either
 259 axis. At $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$ a monoculture needs $\rho^* = 0.596$, which is 12 corrected
 260 firms; at $s = 0.5$ it needs five; at $s = 0.2$ it needs none at all. The regulator's count is $N - \lceil N_c \rceil + 1$
 261 rather than $\lceil \rho^* N \rceil$, and the two differ by a firm exactly when N_c is an integer.

262 **The economic reading.** Correction is a public good: an un-blinded firm captures a private benefit
 263 while the stability it contributes accrues to everyone, so the market free-rides below the threshold



Figure 2: Left: stability against corrected fraction at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$, across efficacies; the gap between the exact threshold and the perfect-correction limit is a result, not an error bar. Right: **the substitution frontier**, the (ρ, s) iso-stability curve, matching at 18 of 18 points. Both are dry runs (Section 8).

and does not reach ρ^* unaided (Bergstrom et al., 1986). Model diversity has the same structure, a firm choosing an idiosyncratic vendor paying privately and supplying systemic stability for free. **The substitution is therefore between two goods that are both under-supplied**, not between a good and a bad.

7 Result 4: the Pigouvian wedge

The third lever is the price instrument. Inside the stable region the binding mode is an AR(1) with coefficient $-m_N$ driven by noise of variance σ^2 , so its stationary variance is $V(m_N) = \sigma^2/(1-m_N^2)$. This is the smooth proxy for divergence risk, and it diverges exactly at the boundary the rest of the paper is about; nothing below uses more than V being positive, increasing and convex. The sign it discards is not lost. Lag-1 autocorrelation is $-m_N$ and therefore negative, so a crowded market's common mode oscillates rather than persists, which is what the supervision estimator of Section 9 reads.

Each firm picks an adaptation aggressiveness a_i , meaning cadence, learning rate or responsiveness, buying concave tracking benefit $B(a_i)$ through $m_i = \mu(a_i)$. Firm i bears weight w_i on common-mode variance and the planner W , with $W > \sum_i w_i$ because every trade has a client bearing execution-quality variance that no dealer's objective contains. That is an assumption, and the results turn on its *sign* and never its size.

Lemma 7 (Marginal crowding share). *At the symmetric point, $\partial m_N / \partial m_i = N_{\text{eff}} v_i^2$ where v is the leading eigenvector of the coupling matrix. The shares sum to N_{eff} rather than to one, and on exchangeable R each equals N_{eff}/N .*

Where a sign condition is needed, and where it is not. Theorem 2's radius needs none. R is positive semidefinite with unit diagonal, so $1 + \kappa(\lambda_i - 1)$ increases in λ_i and the radius sits at $\lambda_{\max}$ whatever the off-diagonal signs. What needs a sign is the *reading* of Lemma 7, that the binding mode is the common one and v is therefore a vector of market shares. **(A5)** supplies it: entrywise nonnegative R puts the leading eigenvector in the nonnegative orthant by Perron-Frobenius, and shared vendors produce positive alignment.

Both readings matter. The *sum* is N_{eff} , so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count of Result 1; the *individual* share $N_{\text{eff}} v_i^2$ collapses to N_{eff}/N on exchangeable R , against a naive commons guess of $1/N$ that is wrong by the factor N_{eff} , which at $N = 20$, $\kappa = 0.8$, $s = 1$ is 16.2.

Theorem 8 (The wedge). *Under Assumption 1 and the welfare assumptions above, at a symmetric profile the private and social first-order conditions are the same equation with a different price on crowding, and the marginal cost a firm ignores is*

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad V'(m_N) = \frac{2\sigma^2 m_N}{(1 - m_N^2)^2}. \quad (6)$$

A per-unit fee t^ on aggressiveness makes the private condition coincide with the social one, so the taxed decentralized equilibrium implements the symmetric social optimum.*

Table 1: Panel coverage and status. Panel 1 is measured in the order-flow simulator rather than in the reference environment, and it reproduces the base result this paper generalizes, so it validates the inherited scaffolding rather than any of the four new results.

Panel	Tests	Status	Outcome
1. Amplification	Section 3	measured	$1.7428 \times, 3.1567 \times$; relative error 0.00
2. (N, s) phase diagram	Thm 2	dry run	48 cells, max error 7.1×10^{-15}
2b. Clustered companion	Thm 2	dry run	measured m_N 1.30; mean index says 0.74
3. Cadence frontier	Thm 4	dry run	175 cells, zero disagreements
4. Herd immunity	Thm 5	dry run	thresholds 12/14/16/20 firms across efficacy
5. Substitution frontier	Thm 5	dry run	18 of 18 points exact
6. Over-adaptation	Thm 8	dry run	12 configurations, zero contradicting Cor. 9

Theorem 8 characterizes the exchangeable symmetric point, which covers the monoculture, orthogonal and supply-chain configurations but switches off the alignment heterogeneity Result 1 exists to represent. Away from it a firm sitting on the crowded mode owes more than its headcount share, and the asymmetric wedge is open (Appendix J). With symmetric weights the private condition ignores the fraction $(N - 1)/N$ of the marginal variance cost borne by other firms, plus the entire client-side exposure. t^* is strictly increasing in N , κ , s and m_N , and diverges at the boundary like $(1 - m_N)^{-2}$. The rate is the economically loaded part: a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to instability.

The provenance channel. m_N depends on s , so the wedge has a second channel: $\partial m_N / \partial s = m_1 \kappa (N - 1)$ against $\partial m_N / \partial a_i = (N_{\text{eff}} / N) \mu'(a_i)$. The same wedge therefore prices **shared-model adoption** directly, which turns the monoculture externality into a fee rather than a warning. The two channels are not symmetric. A firm’s footprint through its own aggressiveness is bounded, since N_{eff} / N falls to κ from above, while the provenance channel is **linear in N and does not decay** against a private gain that does not grow with market size. Unregulated adoption therefore runs toward the unstable configuration, which is why the diversity floor of Section 6 is a different instrument from the cadence cap of Section 5 rather than a restatement of it (Appendix F).

Corollary 9 (Over-adaptation). *The decentralized equilibrium over-adapts relative to the social optimum for every $N \geq 2$, strictly so even with client exposure switched off, and the distortion grows as the market crowds. At $N = 1$ with no client exposure the wedge is exactly zero.*

The fraction the private condition ignores is free of N_{eff} : with no client exposure it is exactly $(N - 1)/N$ whatever κ and s are. Crowding enters the wedge’s *level*, through $V'(m_N)$, and not the distortion’s *shape*, so Corollary 9 is the generic commons result and the provenance asymmetry above is the market-of-models one. Sections 5, 6 and 7 are respectively quantity regulation, a technology mandate or structural remedy, and the price instrument; choosing among them under uncertainty is Weitzman’s question (Weitzman, 1974), and we supply the objects needed to pose it rather than answering it.

8 Experiments

Six panels, one per result, each with its evidential status in its own row of Table 1, because the six are not the same kind of evidence. A panel marked **measured** ran in the order-flow simulator, a genuine N -dealer market with informed flow, spread and inventory that reduces bit for bit to the single-dealer market at $N = 1$. A panel marked **dry run** ran in a linearized reference environment realizing the model’s response geometry and nothing else: it checks that the closed forms were derived from the map we believe they were derived from, which is a check on the derivation rather than evidence about dynamics, and it has caught real errors. It establishes nothing about a market. Every closed form carries a numerical certificate and the harness exits nonzero on any disagreement. Appendix I gives the protocol, which is deterministic from a (config, seed) pair and estimates stability from realized trajectories rather than from an eigensolve. The simulator is described generically pending an artifact release, with code offered to reviewers as anonymized supplementary material.

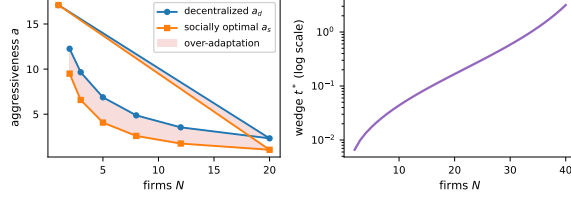


Figure 3: Over-adaptation and the wedge. Left: decentralized against socially optimal aggressiveness across N at $\chi = 1, s = 1$. Right: t^* against N , log scale. Dry run; the boundary divergence fits $(1 - m_N)^{-2}$ at log-log slope -2.000 .

The one measured panel. Panel 1 reproduces a recent preprint’s run in that preprint’s own simulator, at the relative error Table 1 records. Its claim is the monoculture corner $R = \mathbf{11}^\top$, the symmetric law Theorem 2 generalizes, so it is evidence for the scaffolding the four results stand on rather than for any of the four. **The gap to the linear prediction is content, not error:** the market runs -12.9% at $N = 2$ and $+5.2\%$ at $N = 3$, with *opposite signs*, which is what a nonlinear market with saturating flow does around a linearization. The reference environment omits that nonlinearity, which is why it reproduces the prediction exactly. The differential mode measures 3.4×10^{-3} against a theoretical zero at $\kappa = 1$, so the instability is purely common-mode.

Why the rest are not measured. A heterogeneous-response port of the order-flow simulator was built and is exact at flat exposure profiles, but its acceptance gate failed: liquidity and price impact stay shared whatever the response profiles are, leaving a residual larger than several effects these panels resolve, so we drew no figure from it and moved no status. Panel 6 is a dry run for a different reason: the simulator carries no aggressiveness choice variable and no welfare object, so no measured counterpart exists to be built.

Panel 6’s over-adaptation gap widens from 29.0% at $N = 2$ to 119.9% at $N = 20$, with the degenerate $N = 1$ case at exactly zero, as Corollary 9 requires. Its bisection method, which reads m_N from realized dynamics rather than from the closed form, is in Appendix I.

9 Limitations and conclusion

Supervision from public prices. Every quantity above is hidden from the regulator these results address, and near the boundary the system tells on itself. As m_N approaches one, the leading principal component of public quotes has lag-1 autocorrelation approaching one in magnitude, negative because the common mode oscillates, and variance share growing like $1/(1 - m_N)$. A supervisor can therefore estimate distance to instability from public prices alone, and the estimate sharpens as the market nears the boundary it watches for. Consistency, sample complexity and a real-data panel are deferred, the last scoped in the appendix as consistency evidence with a placebo rather than identification.

Limitations. The analysis is linearized around the joint equilibrium (A1), and panel 1 shows what that costs in a real market. **The heterogeneous sweep is closed-form agreement, not measurement,** which is the largest gap in the work; routing liquidity and impact through the response profiles is the identified repair. Theorem 2 is exact for equal moduli (A2) and a two-sided bound otherwise. Share-weighted alignment, the block-secular reduction when (H1) fails, and the sharp concentration rate are named extensions. The $1 - 1/R_0$ collapse assumes the $\kappa = s = 1$ corner, three or more correction levels return a genuine secular equation, and correction that moves a response direction rather than a gain is not modelled. Deployment is synchronous (A3), and heterogeneous clocks break the eigenvector sharing Theorem 4 uses. Pricing instability through stationary variance is itself a modeling choice.

Conclusion. Individual model evaluation cannot certify a market of models. The quantity separating the two questions is the effective number of independent learners, which no evaluation of a single model can see: a firm can pass every test available to it and still be one of the reasons its market fails. Regulation of shared general-purpose models sets thresholds on model scale and ties nothing to a measurable systemic quantity. This paper supplies one, with two substitutable instruments and a price.

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424 A Notation and standing assumptions

425 Symbols marked *new* are this paper’s contribution surface. Everything else is inherited from the base
 426 performative-retraining model and cited rather than re-derived.

Symbol	Meaning	Range
N	firms in the market	integer ≥ 1
d	dimension of each firm’s decision	integer ≥ 1
ε	performative response strength of the environment	> 0
β	sensitivity of the objective to the induced shift	> 0
γ	own-objective curvature	> 0
m_1	single-firm retraining modulus, $\varepsilon\beta/\gamma$	stable iff < 1
κ	spillover between firms through the shared pool	$[0, 1]$
E_i	firm i ’s response Jacobian (<i>new</i>)	$d \times d$
r_{ij}	pairwise alignment of feedback directions (<i>new</i>)	$[-1, 1]$
R	alignment matrix (r_{ij}) (<i>new</i>)	PSD, unit diagonal
$\lambda_{\max}(R)$	effective number of independent learners (<i>new</i>)	$[1, N]$
N_{eff}	effective crowding, $1 + \kappa(\lambda_{\max}(R) - 1)$ (<i>new</i>)	$[1, 1 + \kappa(N - 1)]$
m_N	systemic modulus, $N_{\text{eff}}m_1$ (<i>new</i>)	stable iff < 1
s	shared-model fraction (<i>new</i>)	$[0, 1]$
K	gradient steps per deployment	integer ≥ 1
η	inner learning rate	> 0
c	inner per-step contraction, $1 - \eta\gamma$	$(0, 1)$
γ_{PO}	curvature governing the corrected dynamics	$> \gamma$
θ	γ/γ_{PO} , the correction factor (<i>new</i>)	$(0, 1]$
e	correction efficacy, $1 - \theta$ (<i>new</i>)	$[0, 1]$
N_b	blind (uncorrected) firms (<i>new</i>)	$0 \dots N$
ρ^*	herd-immunity threshold (<i>new</i>)	$[0, 1)$
a_i	firm i ’s adaptation aggressiveness (<i>new</i>)	> 0
t^*	the Pigouvian wedge (<i>new</i>)	

428 The standing assumptions of Assumption 1 are (A1) linearization of the joint retraining map around
 429 the joint equilibrium, with the simulator as the nonlinear check; (A2) equal moduli $m_i = m_1$, under
 430 which Theorem 2 is an identity and outside which it is the two-sided bound of Proposition B.6 below;
 431 (A3) synchronous retraining on a common clock at a common cadence K ; and (A4) coupling through
 432 one pool with a scalar spillover κ . The sign condition (A5), that R is entrywise nonnegative, is not a
 433 standing assumption: Theorem 2’s radius does not need it, and it is invoked only in Section 7, where
 434 the reading of Lemma 7 depends on it. This appendix makes both halves of that statement precise.

435 B The reduction and the alignment spectrum

436 This appendix proves Lemma 1, Theorem 2 and Proposition 3, establishes the properties of R they
 437 use, gives the three anchors, and proves that mean-based diversity indices err with a sign.

438 B.1 Hypotheses and the reduction

439 Firm i deploys $h_i \in \mathbb{R}^d$, and its deployment reshapes the flow it faces through the response Jacobian
 440 $E_i \in \mathbb{R}^{d \times d}$. Write $\text{vec } E_i$ for the d^2 -vector form, $\|\cdot\|_F$ for the Frobenius norm, and $\hat{E}_i = E_i/\|E_i\|_F$
 441 for firm i ’s unit response direction. The reduction uses four hypotheses.

442 **(H1) Rank-one deviation.** Firm i ’s deviation from the joint equilibrium is a scalar x_i along its own
 443 response direction, so the market state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$.

444 **(H2) Equal response magnitude.** $\|E_i\|_F = \varepsilon$ for every i . This is (A2) restated in the new object:
 445 the inherited response strength ε is the Frobenius norm of the response Jacobian.

446 **(H3) Own-channel sensing.** Firm i retrains against the component of the pool distortion lying along
 447 its own response direction, feeling its own contribution in full and each competitor’s with
 448 weight κ .

449 **(H4) Quadratic retraining.** Firm i minimizes a γ -strongly-convex own objective with sensitivity β
 450 to the felt distortion, giving $x_i^+ = -(\beta/\gamma)$ felt $_i$.

451 *Proof of Lemma 1.* The pool carries the aggregate distortion $Z = \sum_j x_j E_j$. By (H3) the distortion
 452 firm i feels is

$$\text{felt}_i = \langle \hat{E}_i, x_i E_i \rangle + \kappa \sum_{j \neq i} \langle \hat{E}_i, x_j E_j \rangle = \varepsilon \left[x_i + \kappa \sum_{j \neq i} r_{ij} x_j \right],$$

453 using $\langle \hat{E}_i, E_j \rangle = \varepsilon r_{ij}$, which is (H2). Since $r_{ii} = 1$ the own term folds into the sum,

$$\text{felt}_i = \varepsilon \left[(1 - \kappa) x_i + \kappa \sum_j r_{ij} x_j \right].$$

454 By (H4), $x^+ = -(\beta/\gamma)\varepsilon[(1 - \kappa)I + \kappa R]x$, and $m_1 = \varepsilon\beta/\gamma$ is the single-firm modulus, which is
 455 (1). $\square$

456 The alignment enters because firm i senses the pool through its own response channel, so two
 457 firms whose deployments perturb the environment in orthogonal directions do not read each other's
 458 perturbations at all, however large those perturbations are. Hypothesis (H3) is what carries that
 459 mechanism. If every E_i is equal then $r_{ij} = 1$ throughout, $R = \mathbf{1}\mathbf{1}^\top$, and the reduction returns the
 460 inherited shared-pool law exactly, so the base model is the corner of the new one in which every
 461 firm's feedback points the same way. Certificate P1 builds random E_i , forms the retraining map from
 462 the felt distortion without ever constructing R , differentiates it numerically and compares against
 463 (1), agreeing to 5.6×10^{-17} at (N, d, κ) equal to $(3, 4, 0.8)$, $(5, 6, 0.35)$, $(8, 5, 1.0)$ and $(6, 7, 0.0)$,
 464 so both spillover endpoints are covered.

465 Hypothesis (H1) is the substantive one. Without it the state is $(h_1, \dots, h_N) \in \mathbb{R}^{Nd}$ and the Jacobian
 466 carries a Kronecker structure with block (i, j) equal to $-(\beta/\gamma)c_{ij}E_j$, where $c_{ij} = 1$ for $i = j$
 467 and κ otherwise, whose spectral radius is not in general $m_1 N_{\text{eff}}$. Reducing that object is the fully
 468 heterogeneous block-secular problem, and it is deferred.

469 B.2 What R is

470 **Lemma B.1.** Let V be the $N \times d^2$ matrix whose rows are the unit vectors $\text{vec } \hat{E}_i$. Then $R = VV^\top$
 471 and: (1) R is positive semidefinite, so every $\lambda_i(R) \geq 0$; (2) $r_{ii} = 1$, hence $\text{tr } R = N$ and
 472 $\sum_i \lambda_i(R) = N$; (3) $|r_{ij}| \leq 1$; (4) $1 \leq \lambda_{\max}(R) \leq N$; (5) $\lambda_{\max}(R) = 1$ if and only if $R = I$; (6)
 473 $\lambda_{\max}(R) = N$ if and only if $R = D\mathbf{1}\mathbf{1}^\top D$ for a sign matrix $D = \text{diag}(\pm 1)$, that is, if and only if
 474 every response is parallel or antiparallel to one common direction.

475 *Proof.* (1) and (2) are immediate from $R = VV^\top$ with unit rows. (3) is Cauchy-Schwarz. For (4),
 476 the eigenvalues are nonnegative and sum to N , so the largest is at least the mean 1 and at most the
 477 total N . For (5), $\lambda_{\max} = 1$ with nonnegative eigenvalues summing to N forces all N eigenvalues to
 478 equal 1. For (6), $\lambda_{\max} = N$ forces every other eigenvalue to zero, so R has rank one, $R = vv^\top$, and
 479 unit diagonal gives $v_i^2 = 1$. $\square$

480 Item (4) is the one the paper leans on: $\lambda_{\max}(R)$ is a genuine effective count, bounded below by one
 481 firm and above by N , with both ends attained.

482 B.3 The spectrum

483 *Proof of Theorem 2.* Write $J = -m_1 B$ with $B = (1 - \kappa)I + \kappa R$. B is symmetric and shares
 484 its eigenvectors with R , with eigenvalues $\nu_i = (1 - \kappa) + \kappa\lambda_i(R)$, so $\rho(J) = m_1 \max_i |\nu_i|$. By
 485 Lemma B.1(1) every $\lambda_i(R) \geq 0$, and $\kappa \leq 1$ gives $1 - \kappa \geq 0$, so every $\nu_i \geq 1 - \kappa \geq 0$. The absolute
 486 value is therefore inert and the maximum sits at $\lambda_{\max}$ rather than at $\lambda_{\min}$:

$$\rho(J) = m_1 [(1 - \kappa) + \kappa\lambda_{\max}(R)] = m_1 [1 + \kappa(\lambda_{\max}(R) - 1)] = m_1 N_{\text{eff}}.$$

487 Stability of the linearized joint map is $\rho(J) < 1$; substituting $m_1 = \varepsilon\beta/\gamma$ and solving for ε gives the
 488 stated form. $\square$

489 Certificate P3 checks this on 384 random configurations with maximum deviation 4.7×10^{-15} .
 490 Dropping the absolute value is legitimate only because R is a Gram matrix. A symmetric unit-
 491 diagonal matrix with entries outside $[-1, 1]$ can put the radius on $\lambda_{\min}$ instead: at $N = 3$ with every

off-diagonal equal to -3 the spectrum is $\{-5, 4, 4\}$ and the negative mode dominates. Cauchy-Schwarz is what rules this out for a real alignment matrix, and P3 certifies both directions.

Corollary B.2. $N_{\text{eff}} \in [1, 1 + \kappa(N - 1)]$, attained at $R = I$ and $R = \mathbf{1}\mathbf{1}^\top$ respectively.

Corollary B.3. $\rho(J) \geq m_1$, with equality if and only if $R = I$. A market of adaptive models sharing an environment is never more stable than one of its members in isolation, and exact neutrality requires fully orthogonal responses.

Corollary B.3 says the externality has a sign: coupling can only amplify feedback, never damp it, so there is no configuration in which crowding does a market a favour. Individual modes can be more damped than m_1 , and the differential mode at $m_1(1 - \kappa)$ is one, but the radius governs and the radius cannot fall below m_1 .

B.4 Which mode binds

Proposition B.4. If R is entrywise nonnegative and irreducible then $B = (1 - \kappa)I + \kappa R$ is entrywise nonnegative and irreducible, so by Perron-Frobenius its leading eigenvector is strictly positive and simple, and the binding mode is a genuine common mode in which every firm deviates in the same direction.

Entrywise nonnegativity of R is exactly the condition that no firm's feedback anti-aligns with another's. Shared vendors and shared pretraining corpora produce positive alignment, so the realistic regime satisfies it, and the paper states the condition rather than assuming it silently. This is (A5), and Proposition B.4 is the only place it is used.

B.5 Three anchors, and a mode swap

Monoculture. $R = \mathbf{1}\mathbf{1}^\top$ has rank one with eigenvalue N on the all-ones direction, so $\lambda_{\max} = N$ and $N_{\text{eff}} = 1 + \kappa(N - 1)$. This is the inherited law whose predicted $2\times$ and $3\times$ at $N = 2, 3$ were measured at $1.74\times$ and $3.16\times$.

Orthogonal. $R = I$ gives $\lambda_{\max} = 1$, $N_{\text{eff}} = 1$ and $m_N = m_1$. A hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm.

Simplex. With $r_{ij} = -1/(N - 1)$ off the diagonal,

$$R_{\text{simplex}} = \frac{N}{N-1}I - \frac{1}{N-1}\mathbf{1}\mathbf{1}^\top,$$

whose diagonal is $N/(N - 1) - 1/(N - 1) = 1$ as required. Since $\mathbf{1}\mathbf{1}^\top$ has eigenvalue N on the all-ones direction and 0 on its complement, the spectrum is 0 on $\mathbf{1}$ and $N/(N - 1)$ with multiplicity $N - 1$. So $\lambda_{\max} = N/(N - 1)$, $N_{\text{eff}} = 1 + \kappa/(N - 1)$, and the radius tends to m_1 as N grows. The value $m_1(1 - \kappa)$ does sit in the spectrum, carried by the all-ones direction at $\lambda = 0$, and it matches the inherited differential-mode eigenvalue derived by a different route, but it is not the spectral radius.

Under the simplex the all-ones direction carries the *smallest* eigenvalue, so the leading eigenvector is orthogonal to it and (A5) fails. That is consistent with Proposition B.4, since the simplex has negative entries and Perron-Frobenius does not apply. Certificate P4 checks all three parts, that the all-ones direction is annihilated, that the leading eigenvector is orthogonal to it, and that the entries are negative, at $N \in \{3, 5, 10, 30, 50\}$.

B.6 A witness for strict refinement

Proof of Proposition 3. Both markets satisfy (H1) to (H4) with the same response magnitude ε , the same objective curvature γ and sensitivity β , and the same spillover κ . They therefore share $m_1 = \varepsilon\beta/\gamma$ and share the joint map (1) in everything except R , so any quantity computed from response magnitudes, objective curvature and spillover alone takes the same value in both. By Theorem 2 the radii differ: $\lambda_{\max}(I) = 1$ gives $m_N = m_1$, while $\lambda_{\max}(\mathbf{1}\mathbf{1}^\top) = N$ gives $m_N = m_1(1 + \kappa(N - 1))$. The hypothesis $1/(1 + \kappa(N - 1)) < m_1 < 1$ puts the first below one and the second above it. The two markets thus agree on every scalar of the stated kind and disagree in stability, so a criterion that is a function of those scalars alone returns one verdict for both and is wrong about at least one. The interval is nonempty for every $N \geq 2$ and $\kappa > 0$. $\square$

538 The witness is not a knife-edge construction. Any m_1 in an interval of length $1 - 1/N_{\text{eff}}$ works,
 539 which at $N = 20$ and $\kappa = 0.8$ is 94% of the unit interval.

540 B.7 Mean-alignment indices err, and always in the same direction

541 **Proposition B.5.** Let $\bar{m}$ be the off-diagonal mean of R . Then $\lambda_{\max}(R) \geq 1 + (N - 1)\bar{m}$, hence
 542 $N_{\text{eff}} \geq 1 + \kappa(N - 1)\bar{m}$, with equality if and only if $\mathbf{1}$ is a leading eigenvector of R , that is, if and
 543 only if R has constant row sums.

544 *Proof.* $\mathbf{1}^\top R \mathbf{1} = N + N(N - 1)\bar{m}$, since the diagonal contributes N . The Rayleigh quotient at the
 545 all-ones vector gives $\lambda_{\max}(R) \geq \mathbf{1}^\top R \mathbf{1} / N = 1 + (N - 1)\bar{m}$, with equality if and only if $\mathbf{1}$ attains
 546 the Rayleigh maximum, that is if and only if $R \mathbf{1}$ is parallel to $\mathbf{1}$. Multiplying by κ and adding $1 - \kappa$
 547 preserves the inequality. $\square$

548 Certificate P5 finds zero violations on 360 random R and confirms equality on uniform R . Two
 549 readings follow. First, a mean-similarity diversity index never over-states systemic risk: it under-states
 550 it or is exact, so every error it makes is in the unsafe direction. A regulator using mean similarity
 551 is not merely imprecise, it is imprecise in the direction that lets unstable markets pass. Second,
 552 $\mathbf{1}^\top R \mathbf{1} = \|\sum_i \text{vec } \hat{E}_i\|^2 \geq 0$ forces $\bar{m} \geq -1/(N - 1)$, attained exactly at the simplex.

553 The gap is largest at the realistic topology, which is the vendor with a plurality rather than a monopoly.
 554 Take $N = 10$ with three firms fully aligned and every other pair orthogonal, so that R is block
 555 diagonal with a 3×3 all-ones block and $\lambda_{\max} = 3$. Three of the forty-five pairs are aligned, so
 556 $\bar{m} = 1/15$ and the mean index reports $\lambda_{\max} = 1.6$. At $\kappa = 0.8$ the true N_{eff} is 2.60 against the
 557 index's 1.48, an understatement by a factor of 1.757, and a market at $m_1 = 0.5$ then runs $m_N = 1.30$
 558 and is unstable while the index reports 0.74 and calls it safe with margin.

559 The failure is not only one of level. The simplex *minimizes* the mean at $\bar{m} = -1/(N - 1)$ while
 560 carrying $\lambda_{\max} = N/(N - 1)$, which exceeds one, so it is spectrally worse than orthogonality even
 561 though it has the smaller mean. The mean therefore does not order configurations correctly, which is
 562 why every stability claim in this paper uses the spectral form and never a mean.

563 B.8 Heterogeneous moduli

564 **Proposition B.6.** Let $M = \text{diag}(m_i)$ with every $m_i > 0$, and let $J = -M[(1 - \kappa)I + \kappa R]$. Then

$$\max_i m_i \leq \rho(J) \leq \left(\max_i m_i\right) N_{\text{eff}}.$$

565 *Proof.* J is not symmetric, but the congruence $S = M^{1/2}$ gives $M^{1/2}(MB)M^{-1/2} =$
 566 $M^{1/2}BM^{1/2}$, so $\rho(J) = \lambda_{\max}(A)$ with $A = (1 - \kappa)M + \kappa M^{1/2}RM^{1/2}$, symmetric and pos-
 567 itive semidefinite. For the lower bound, $e_i^\top A e_i = (1 - \kappa)m_i + \kappa m_i r_{ii} = m_i$ since $r_{ii} = 1$, and the
 568 Rayleigh quotient at each e_i gives $\lambda_{\max}(A) \geq m_i$ for every i . For the upper bound, $R \preceq \lambda_{\max}(R)I$
 569 in the Loewner order and congruence by $M^{1/2}$ preserves it, so $M^{1/2}RM^{1/2} \preceq \lambda_{\max}(R)M$ and
 570 hence $A \preceq N_{\text{eff}}M$, giving $\lambda_{\max}(A) \leq N_{\text{eff}} \max_i m_i$. $\square$

571 The bound is exact at $\kappa = 0$ and at $R = I$, where $A = M$ and both sides equal $\max_i m_i$, and its upper
 572 half is exact at equal moduli, where it is Theorem 2. A mean-modulus formula is provably wrong:
 573 at $R = I$ the firms decouple and $\rho(J) = \max_i m_i$, certified with m ranging over $\{0.1, \dots, 0.9\}$ at
 574 $\rho = 0.900$ against a mean of 0.400. Certificate P8 finds no violation on 300 random draws.

575 C Concentration to the supply-chain limit

576 The decomposition $E_i = \sqrt{s} E_{\text{shared}} + \sqrt{1 - s} \Xi_i$ is meant to deliver $R \rightarrow (1 - s)I + s \mathbf{1} \mathbf{1}^\top$ and
 577 therefore $N_{\text{eff}} \rightarrow 1 + \kappa s(N - 1)$. Saying the cross terms vanish in expectation is not enough for a
 578 stability claim, because a market is unstable or not at the realized alignment matrix rather than at its
 579 mean.

(D) **Design assumption.** E_{shared} and the Ξ_i have independent entries, mean zero, unit variance, sub-Gaussian with a common constant, and the Ξ_i are independent of each other and of E_{shared} .

This is an idealization. Real response Jacobians are neither Gaussian nor independent across firms beyond the shared component, and what the argument uses is sub-Gaussian entries and independence of the idiosyncratic parts. Write $D = d^2$, $u = \text{vec } E_{\text{shared}}$ and $\xi_i = \text{vec } \Xi_i$, so that $\text{vec } E_i = \sqrt{s}u + \sqrt{1-s}\xi_i$ in $\mathbb{R}^D$. Under (D), $\|u\|^2/D$ and $\|\xi_i\|^2/D$ concentrate at 1 while $u^\top \xi_j/D$ and $\xi_i^\top \xi_j/D$ concentrate at 0, all at rate $1/\sqrt{D}$, so $r_{ij} \rightarrow s$ for $i \neq j$ and $\lambda_{\max}(R_\infty) = 1 + s(N-1)$ exactly by the rank-one structure.

Lemma C.1. Under (D) there is an absolute constant C such that for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_{i \neq j} |r_{ij} - s| \leq C \frac{\sqrt{\log(N/\delta)}}{d}.$$

Proof. Each of $\|u\|^2/D - 1$, $\|\xi_i\|^2/D - 1$, $u^\top \xi_i/D$ and $\xi_i^\top \xi_j/D$ is an average of D independent mean-zero sub-exponential terms, so Bernstein's inequality gives a tail $2 \exp(-cDt^2)$ for $t \in (0, 1)$. There are $O(N^2)$ such quantities across all pairs, and a union bound with $t = C\sqrt{\log(N/\delta)/D}$ controls all of them simultaneously with probability at least $1 - \delta$. The ratio defining r_{ij} is a smooth function of these quantities with bounded derivatives in a neighbourhood of the limit point, so the same rate carries to $r_{ij} - s$, and $\sqrt{D} = d$ gives the statement. $\square$

The inner products average over d^2 matrix entries, so the fluctuation is $1/d$. Certificate P7 measures a log-log slope of -1.018 against a target of -1 , at $N = 8$, $s = 0.6$, $d \in \{8, 16, 32, 64, 128\}$ and 40 trials per point.

Proposition C.2. Under the event of Lemma C.1,

$$\begin{aligned} |\lambda_{\max}(R) - (1 + s(N-1))| &\leq C \frac{N\sqrt{\log(N/\delta)}}{d}, \\ |N_{\text{eff}} - (1 + \kappa s(N-1))| &\leq \kappa C \frac{N\sqrt{\log(N/\delta)}}{d}. \end{aligned}$$

Proof. Write $\Delta = R - R_\infty$, which has zero diagonal since both matrices have unit diagonal. Weyl's inequality gives $|\lambda_{\max}(R) - \lambda_{\max}(R_\infty)| \leq \|\Delta\|_2$, and $\|\Delta\|_2 \leq \|\Delta\|_F \leq \sqrt{N(N-1)} \max_{i \neq j} |\Delta_{ij}| \leq N \max_{i \neq j} |r_{ij} - s|$. Apply Lemma C.1; the N_{eff} form follows on multiplying by κ . $\square$

The bound matters in relative terms, because N_{eff} itself grows with N . For large N the limit is $1 + s(N-1) \approx sN$, so the relative error is of order $C\sqrt{\log N}/(sd)$, with the N in the numerator cancelling against the N in N_{eff} . The closed form is therefore accurate once $d \gg \sqrt{\log N}/s$, a condition on dimension against vendor concentration rather than against firm count, and it degrades as $s \rightarrow 0$, which is correct: at low shared-model fraction the limit matrix is close to the identity and the leading eigenvalue is set by fluctuations rather than by the shared component. In the instantiation the base model calibrates, d is in the low hundreds and N in the tens, so the requirement is met with room for s above roughly 0.2.

The step $\|\Delta\|_2 \leq \|\Delta\|_F$ is crude and is used because it is elementary and unconditional. For a Δ with the Wishart-type structure that (D) produces, the operator norm concentrates at $\sqrt{N}/d$ rather than at N/d , which would improve the relative error to $1/(sd\sqrt{N})$, better for larger markets rather than worse. The sharp rate is reported as measured and not as proved, since claiming it otherwise would violate the paper's rule on status flags.

D The cadence composition

Lemma D.1. Let firm i retrain by gradient descent on a γ -strongly-convex quadratic own objective whose minimizer is the frozen best response b , with step size η . Then K steps from z_0 land at $z_K = b + c^K(z_0 - b)$ with $c = 1 - \eta\gamma$.

622 *Proof.* The gradient is $\gamma(z - b)$, so the update is $z \mapsto z - \eta\gamma(z - b) = b + (1 - \eta\gamma)(z - b)$. Induct
623 on K . $\square$

624 Here $c \in (0, 1)$ requires $0 < \eta < 2/\gamma$, and $\eta \leq 1/\gamma$ gives $c \in [0, 1]$. Certificate Q1 runs
625 actual gradient descent and measures the realized gap ratio, exact to 10^{-12} at three (η, γ) pairs and
626 $K \in \{1, 3, 7, 20\}$. The constant c is never written into the loop; it is whatever the loop does.

627 **Lemma D.2.** For a single firm whose full-retraining best response has slope $-m$, the K -step outer
628 map has slope $\mu(K) = -m + c^K(1 + m)$.

629 *Proof.* The frozen best response at the current deployment h_t is $b = -mh_t$, so by Lemma D.1,
630 $h_{t+1} = -mh_t + c^K(h_t + mh_t) = h_t[-m + c^K(1 + m)]$. $\square$

631 At $K \rightarrow \infty$ this gives $-m$, the full-retraining slope, and at $K = 0$ it gives the identity, since a firm
632 taking no gradient steps does not move. The inherited single-firm lemma is therefore reproduced
633 rather than assumed.

634 The composition rests on one hypothesis, stated so that it can be checked.

635 (C) The contraction c is the same scalar in the joint market as in the single-firm market. It is built
636 from own-objective curvature γ and inner step size η , neither of which involves N , κ or R :
637 competitors change *where* firm i is heading, through the frozen best response, and not *how*
638 *fast* it gets there.

639 This is the only place the composition can fail, so it is certified rather than argued. Certificate C12
640 measures the realized one-step contraction inside the joint market across 45 configurations spanning
641 $N \in \{1, 2, 5, 12, 30\}$, $\kappa \in \{0, 0.5, 1\}$ and $m_1 \in \{0.05, 0.15, 0.6\}$ with random R at each, and the
642 spread across all of them is 4.4×10^{-16} .

643 *Proof of Theorem 4.* All firms compute their frozen best responses from the common current state x ,
644 giving $b = Jx$, then each takes K inner steps. By Lemma D.1 applied coordinatewise, and using (C)
645 so that the same c applies in every coordinate,

$$x_{t+1} = Jx_t + c^K(x_t - Jx_t) = [c^K I + (1 - c^K)J]x_t.$$

646 This matrix is an affine function of J , so it shares J 's eigenvectors, with eigenvalues $\mu_i(K) =$
647 $c^K - (1 - c^K)m_1\nu_i$ where $\nu_i = (1 - \kappa) + \kappa\lambda_i(R)$. On the mode carrying $\lambda_{\max}(R)$, where
648 $m_1\nu_{\max} = m_N$ by Theorem 2, this is $\mu_N(K) = c^K - (1 - c^K)m_N = -m_N + c^K(1 + m_N)$.

649 For the stability claim, $\mu_i(K)$ is strictly decreasing in ν_i because $(1 - c^K)m_1 > 0$ for every $K \geq 1$.
650 Every $\nu_i \geq 0$ by Lemma B.1(1) with $\kappa \leq 1$, so $\mu_i(K) \leq c^K < 1$ for all i and the upper side never
651 binds, whatever R is; and $\min_i \mu_i(K)$ is attained at $\nu_{\max}$, so the constraint is exactly $\mu_N(K) > -1$.
652 Expanding,

$$c^K - (1 - c^K)m_N > -1 \iff m_N < \frac{1 + c^K}{1 - c^K} \iff c^K > \frac{m_N - 1}{m_N + 1},$$

653 and taking logarithms, with $\ln c < 0$ flipping the inequality on division, gives $K < K_{\max}$. When
654 $m_N \leq 1$ the right-hand side of the last inequality is non-positive while $c^K > 0$, so the constraint is
655 vacuous and every cadence is stable. $\square$

656 Certificate Q2 builds the joint map by running the actual inner gradient descent on each basis vector
657 and comparing against $c^K I + (1 - c^K)J$ at 32 combinations of (N, κ, m_1, K) , agreeing below
658 10^{-12} ; C8 checks the frontier on 51600 integer cases with zero mismatches; and Q5 iterates the
659 actual joint map to convergence or divergence on 59 decisive trials at random (N, κ, s, m_1, K) , with
660 zero disagreements with the predicted side.

661 The monotonicity argument replaces an earlier route through the size of the differential modes. It is
662 shorter and it is unconditional: it needs no Perron-Frobenius condition, because monotonicity places
663 the extreme at $\lambda_{\max}$ regardless of the sign pattern of R .

664 **Corollary D.3.** $K_{\max}$ is strictly decreasing in m_N , and hence in s .

665 *Proof.* With $f(m) = \ln \frac{m-1}{m+1}$ we have $f'(m) = \frac{1}{m-1} - \frac{1}{m+1} = \frac{2}{m^2-1} > 0$ for $m > 1$, and dividing
 666 by $\ln c < 0$ flips the sign. Since $m_N = m_1(1 + \kappa s(N-1))$ is increasing in s , the composition is
 667 decreasing in s . $\square$

668 Two readings the body uses. Since K is a positive integer, the realized window is $\lfloor K_{\max} \rfloor$ while the
 669 plotted frontier is continuous, and certificate Q4 confirms that $\lfloor K_{\max} \rfloor$ is the largest stable integer and
 670 $\lfloor K_{\max} \rfloor + 1$ is unstable at five (c, m_N) pairs. And a feasible cadence exists if and only if $K_{\max} > 1$,
 671 which is exactly $m_N < (1+c)/(1-c)$: past that level of effective crowding the market is unstable
 672 at every retraining frequency, and minimum-cadence operation multiplies the sustainable crowding
 673 by exactly $(1+c)/(1-c)$, a factor of 9 at $c = 0.8$. Critical crowding is therefore not a separate
 674 result but the $K = 1$ instance of the frontier.

675 Worked at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$ and $c = 0.8$, so that $N_{\text{eff}} = 1 + 23.2s$:

	s	N_{eff}	m_N	$K_{\max}$
676	0.25	6.80	1.020	20.68
	0.50	12.60	1.890	5.28
	1.00	24.20	3.630	2.53

677 All three m_N values sit below the critical crowding level of 9, which is why every column has a
 678 finite window. The table is not reproducible without c , so $c = 0.8$ is stated wherever it appears. The
 679 operational reading is that the externality is not only that a competitor's entry consumes a firm's
 680 retraining budget, but that a competitor's choice of vendor does too, without their entering at all and
 681 without either firm doing anything wrong.

682 Two scope notes close this appendix. By Lemma D.1 the deployed model's gap to the frozen best
 683 response after a round is exactly c^K times the gap before it, so c^K is simultaneously what buys stability
 684 and what measures the staleness of the deployed model: the lever is not free, and its price is quantified
 685 in the same parameter that delivers its benefit. And at $m_N = 1$ exactly, $\mu_N(K) = -1 + 2c^K$ lies
 686 strictly inside $(-1, 1)$ for every finite K , so lazy retraining strictly stabilizes the marginally unstable
 687 market, with a margin $2c^K$ that decays geometrically and falls below double precision past $K = 171$
 688 at $c = 0.8$. That last is a floating-point limit and not a statement about markets, and the certificate
 689 records it as such.

690 E The mixed market

691 Of N firms sharing one pool, N_b are blind and run the ordinary retraining loop while $N_{\text{corr}} = N - N_b$
 692 are corrected and run the feedback-aware update governed by γ_{PO} . Write $\theta = \gamma/\gamma_{PO} \in (0, 1]$, so
 693 that $\theta = 1$ is no correction and $\theta \rightarrow 0$ is the strong-correction limit.

694 **(M)** Correction scales a firm's modulus: a corrected firm carries θm_1 where a blind firm carries
 695 m_1 . This is the faithful reading of the single-firm result, in which the corrected loop's
 696 modulus is $m_1\gamma/\gamma_{PO}$. Correction changes a firm's gain and not its response direction, so R
 697 is untouched.

698 Take the supply-chain alignment $R = (1-s)I + s\mathbf{1}\mathbf{1}^\top$ and write $k_s = \kappa s$, so that $B = (1 -$
 699 $k_s)I + k_s\mathbf{1}\mathbf{1}^\top$. By Proposition B.6 the radius of $J = -MB$ is $\lambda_{\max}(A)$ for $A = M^{1/2}BM^{1/2}$, and
 700 substituting B ,

$$A = (1 - k_s)M + k_s w w^\top, \quad w = M^{1/2}\mathbf{1}, \quad w_i = \sqrt{m_i}.$$

701 So A is a diagonal matrix plus a positive-weight rank-one term, and its eigenvalues not equal to
 702 a diagonal entry solve the secular equation $1 + k_s \sum_i m_i / ((1 - k_s)m_i - \lambda) = 0$. With only two
 703 distinct moduli the sum collapses to two terms and the equation clears to a quadratic, which is why
 704 the mixed market has a closed form at all and why three or more correction levels would not.

705 *Proof of Theorem 5.* Write $a = (1 - k_s)m_1$ and $b = (1 - k_s)\theta m_1$ for the two diagonal values.
 706 Clearing denominators in the secular equation gives

$$(a - \lambda)(b - \lambda) + k_s N_b m_1 (b - \lambda) + k_s N_{\text{corr}} \theta m_1 (a - \lambda) = 0,$$

707 that is $\lambda^2 - P\lambda + Q = 0$ with

$$P = a + b + k_s m_1 (N_b + \theta N_{\text{corr}}), \quad Q = \theta m_1^2 (1 - k_s) (1 + k_s (N - 1)) = \theta m_1^2 (1 - k_s) N_{\text{eff}}.$$

708 The two secular roots are this quadratic's roots, and the eigenvectors constant within each block
709 and orthogonal to w supply the rest of the spectrum, namely a with multiplicity $N_b - 1$ and b with
710 multiplicity $N_{\text{corr}} - 1$. Because the rank-one weight k_s is positive, the eigenvalues of A interlace
711 those of $(1 - k_s)M$, so the largest eigenvalue of A exceeds every diagonal entry and is therefore the
712 larger secular root rather than a degenerate one, giving $\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q})$. $\square$

713 Certificate H1 reports a maximum error of 2.5×10^{-14} against dense eigensolves over 6000 random
714 draws. The appearance of N_{eff} inside Q is not cosmetic: the mixed market inherits Theorem 2's
715 crowding through that product, which is why every statement here carries s .

716 **Degenerate blocks.** When $N_b = 0$ or $N_b = N$ one block is empty, but the quadratic still carries that
717 block's factor and leaves a phantom root behind. At $N_b = 0$ the naive quadratic returns $(1 - k_s)m_1$, a
718 value no firm's modulus supports, and at small θ that phantom exceeds the true radius: 0.132 against
719 a true 0.043 at $N = 12$, $m_1 = 0.3$, $\kappa = 0.8$, $s = 0.7$ and $\theta = 0.02$. The correct values are the
720 single-block ones, $\rho = m_1 N_{\text{eff}}$ at $N_b = N$ and $\rho = \theta m_1 N_{\text{eff}}$ at $N_b = 0$, and any implementation
721 must branch on this. Certified in H2.

722 **Both limits are corollaries.** At $\theta = 1$, $P = m_1[2(1 - k_s) + k_s N]$ and $Q = m_1^2(1 - k_s)N_{\text{eff}}$,
723 whose roots are $m_1 N_{\text{eff}}$ and $m_1(1 - k_s)$, recovering Theorem 2 exactly. As $\theta \rightarrow 0$, $Q \rightarrow 0$, so
724 one root goes to zero and the other to $P \rightarrow m_1(1 + \kappa s(N_b - 1))$, the blind-block condition. The
725 argument that corrected firms transmit no feedback and drop out of the cycle is right, and it is now a
726 corollary of the exact root rather than a separate claim.

727 E.1 The limit errs, and it errs optimistic

728 **Proposition E.1.** $\rho(J)$ is nondecreasing in θ .

729 *Proof.* $A_{ij} = \sqrt{m_i m_j} B_{ij}$ and B is entrywise nonnegative for $k_s \in [0, 1]$, so A is entrywise
730 nonnegative. Every m_i is nondecreasing in θ , constant for blind firms and θm_1 for corrected ones, so
731 every entry of A is nondecreasing in θ . Since A is symmetric and nonnegative, by Perron-Frobenius
732 its Rayleigh maximizer may be taken in the nonnegative orthant, and for such an x the form $x^\top A(\theta)x$
733 is nondecreasing in θ . Taking x to be the maximizer at $\theta_1 < \theta_2$,

$$\lambda_{\max}(A(\theta_2)) \geq x^\top A(\theta_2)x \geq x^\top A(\theta_1)x = \lambda_{\max}(A(\theta_1)). \quad \square$$

734 Two readings follow and they point opposite ways. The reassuring one is that correction never
735 backfires: more correction is always weakly stabilizing, so there is no perverse configuration in
736 which un-blinding a firm destabilizes the market. The uncomfortable one is that $\rho(\theta) \geq \rho(0)$, so the
737 strong-correction limit *under-states* the radius. The limit is optimistic rather than conservative, and it
738 can call a market stable that is unstable at any finite correction strength. On uniform random draws
739 with $\theta \in [0.01, 1]$ the limit calls 11.8% of configurations stable that are not. Three worked cases, all
740 of which the limit reads as comfortably stable:

	Configuration	limit reads	truly unstable unless
741	$N = 10, N_b = 6, m_1 = 0.15, \kappa = 0.8, s = 1$	0.750	$\gamma_{PO} > 1.9\gamma$
	$N = 30, N_b = 8, m_1 = 0.10, \kappa = 0.9, s = 0.8$	0.604	$\gamma_{PO} > 3.9\gamma$
	$N = 20, N_b = 10, m_1 = 0.12, \kappa = 0.8, s = 1$	0.984	$\gamma_{PO} > 58.6\gamma$

742 Certificate C18 finds zero violations of Proposition E.1 on 3000 random configurations, each swept
743 over 50 values of θ , and reports the verdict flips above. This is why the paper states the exact
744 two-block root as the theorem and the limit as a corollary carrying its own error direction: shipping
745 the limit alone would state a stability criterion that errs in the unsafe direction without saying so.

746 E.2 Imperfect correction

747 *Proof of the imperfect-correction law and of Corollary 6.* At $\kappa = s = 1$ the quadratic degenerates,
 748 since $1 - k_s = 0$ kills a , b and Q , leaving $\rho = P = m_1(N_b + \theta N_{\text{corr}})$. Writing the corrected fraction
 749 as $\rho_c = 1 - N_b/N$ and solving $\rho < 1$ for it gives

$$\rho^*(\theta) = \frac{1 - 1/m_N}{1 - \theta} = \frac{1 - 1/m_N}{e}, \quad e = 1 - \theta = 1 - \gamma/\gamma_{PO},$$

750 which is the standard epidemiological coverage requirement for an imperfect vaccine, with the clean
 751 law $1 - 1/m_N$ as its perfect-efficacy corner. The required fraction exceeds one, so that no corrected
 752 fraction stabilizes the market, exactly when $\theta > 1/m_N$, that is when $\gamma_{PO}/\gamma \leq m_N$, which is
 753 Corollary 6. $\square$

754 Certificate H5 reports agreement to 1.8×10^{-15} over 120 configurations, with the radius form checked
 755 against dense eigensolves, and H6 certifies the critical efficacy. At $m_N = 2.5$ the corrected update
 756 must deliver at least a $2.5 \times$ curvature improvement or the market cannot be stabilized by un-blinding
 757 at all, whatever fraction is treated. This is the exact structural parallel of the critical crowding level of
 758 Appendix D: each lever has a regime past which it stops working, and stating both is what makes the
 759 substitution frontier an honest object rather than an extrapolation.

760 The correspondence is worth being precise about. It now transfers not only the threshold but the
 761 standard refinement of the threshold, which is what a structural analogy predicts and a decorative one
 762 does not. Away from $\kappa = s = 1$ the closed form is no longer exact and the exact threshold comes
 763 from setting the quadratic's larger root to one; the law remains accurate to well under one firm's
 764 granularity at the parameters this paper uses.

765 E.3 The threshold in firms

766 **Proposition E.2.** The strong-correction limit is stable if and only if $N_b < N_c(s) = 1 + (1/m_1 -$
 767 $1)/(\kappa s)$.

768 *Proof.* Rearrange $m_1(1 + \kappa s(N_b - 1)) < 1$. $\square$

769 Certificate C14 finds zero mismatches on 4000 draws for this form. The clamped restatement is
 770 *not* equivalent: writing $\rho_c = 1 - N_b/N$ and $\rho^* = \max(0, 1 - N_c(s)/N)$, the criterion $\rho_c > \rho^*$
 771 fails on 134 of those same 4000 draws, every failure at the same corner, namely $N_b = N$ with
 772 $N_c(s) \geq N$, the all-blind market that is stable because it needs no correction at all and that a clamp
 773 at zero combined with a strict inequality excludes. The paper therefore states $N_b < N_c(s)$ as the
 774 theorem, which is primitive and exact, and presents the clamped ρ^* as the policy object, read as the
 775 fraction that must be corrected, with $\rho^* = 0$ meaning no requirement rather than a strict inequality to
 776 clear.

777 Since ρ^*N is generally not an integer and the panels run at N in the tens, where one firm is several
 778 percentage points, the realized requirement is

$$\text{minimum corrected firms} = N - \lceil N_c(s) \rceil + 1,$$

779 clamped to $[0, N]$, because the largest stable blind count is $\lceil N_c \rceil - 1$. The natural guess $\lceil \rho^*N \rceil$
 780 agrees on every random draw, since random draws never land on an exact integer N_c , but it is off by
 781 one exactly when N_c is an integer: at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and the s giving $N_c = 8$, the
 782 truth is 13 corrected firms and the guess says 12. Certificate H4 checks the correct form exactly on
 783 3000 draws. Worked at $N = 20$, $m_1 = 0.15$ and $\kappa = 0.8$:

s	$N_c(s)$	ρ^*	minimum corrected firms
1.0	8.08	0.596	12
0.5	15.17	0.242	5
0.2	36.42	0	0

785 At $\kappa = s = 1$ we have $N_c = 1/m_1$ and $N_{\text{eff}} = N$, so $\rho^* = 1 - 1/(m_1N) = 1 - 1/m_N$, the
 786 epidemiological threshold with the systemic modulus as the reproduction number: a market of ten

787 firms at $m_N = 2.5$ needs 60% un-blinded. And ρ^* is increasing in s , strictly so wherever it is positive,
 788 which is what makes model diversity and corrected learning substitutes and what the (ρ, s) frontier
 789 plots. Certified in C15 and C16.

790 Panel 4 runs at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$ and $s = 1$, where the imperfect-correction law is
 791 not exact, and measures thresholds of 12, 14, 16 and 20 firms at efficacies 1.00, 0.90, 0.75 and 0.60
 792 against a law predicting 12, 14, 16 and 20. That is four out of four at the integer granularity a market
 793 actually faces. It is a dry run in the linearized reference environment and is recorded as evidence
 794 rather than as proof; the exact statement away from $\kappa = s = 1$ remains the quadratic's root.

795 **F The Pigouvian wedge**

796 **F.1 The welfare object**

797 Drive the linearized joint map with noise. Under (A1) and (A5) the binding mode is the common
 798 one, whose deviation z_t from the joint equilibrium obeys $z_{t+1} = -m_N z_t + \xi_t$ with ξ_t independent,
 799 mean zero and variance σ^2 . This is an AR(1) with coefficient $-m_N$. Stationarity requires $|m_N| < 1$,
 800 which is the stability condition the rest of the paper works with, and taking variances of a stationary
 801 solution gives $V = m_N^2 V + \sigma^2$, hence

$$V(m_N) = \frac{\sigma^2}{1 - m_N^2}.$$

802 The coefficient is $-m_N$ and not m_N because the retraining Jacobian is $J = -m_1[(1 - \kappa)I + \kappa R]$ and
 803 the common mode inherits its sign. The stationary variance is blind to that sign, since the recursion
 804 enters squared, which is why the same expression serves both. The sign is not lost information: it
 805 reappears in the lag-1 autocorrelation, which is $-m_N$ and therefore negative, so a crowded market
 806 shows oscillatory rather than persistent common-mode error. That is the observable the supervision
 807 estimator reads, and it is why a positive-autocorrelation diagnostic would look for the wrong thing.
 808 Certificate C19 checks both facts by simulation rather than asserting them.

809 V is the paper's price of instability, and it is a modeling choice: stationary variance is the smooth
 810 proxy for divergence risk, finite everywhere strictly inside the stable region and divergent exactly at
 811 the boundary the rest of the paper is about. Nothing below depends on the functional form beyond V
 812 being positive, increasing and convex in m_N on $[0, 1)$, all three of which the closed form satisfies.

813 **(W1) Aggressiveness.** Each firm chooses $a_i > 0$, meaning cadence, learning rate or responsiveness:
 814 anything that scales its own modulus. Write $m_i = \mu(a_i)$ with μ differentiable, strictly
 815 increasing and common across firms. Aggressiveness moves a firm's gain and not its
 816 response direction, so R and κ are held fixed.

817 **(W2) Benefit.** Aggressiveness buys tracking benefit $B(a_i)$, strictly increasing and strictly concave.
 818 Only firm i 's own benefit depends on a_i , so the entire interaction runs through m_N .

819 **(W3) Exchangeable alignment.** R is invariant under a common permutation of firms. The supply-
 820 chain alignment is exchangeable, as are the monoculture and orthogonal corners. This buys
 821 the uniform leading eigenvector at the symmetric point.

822 **(W4) Client exposure.** The planner's variance weight satisfies $W > \sum_i w_i$, with all weights posi-
 823 tive.

824 (W4) is stated as an assumption and is not microfounded. Every trade has a client on the other side
 825 who bears execution-quality variance while no dealer's objective contains it, so the planner's weight
 826 on common-mode variance strictly exceeds what the dealers together internalize. The results below
 827 turn on the sign of $W - \sum_i w_i$ and never on its size, so a microfoundation would fix a magnitude the
 828 theorem does not use.

829 **F.2 The marginal crowding share**

830 *Proof of Lemma 7.* By Proposition B.6, for $M = \text{diag}(m_1, \dots, m_N)$ and $B = (1 - \kappa)I + \kappa R$ the ra-
 831 dius of $J = -MB$ is $\lambda_{\max}(A)$ for $A(M) = M^{1/2} B M^{1/2}$. Since $dM^{1/2}/dm_i = (1/(2\sqrt{m_i}))e_i e_i^\top$,

$$\frac{dA}{dm_i} = \frac{1}{2\sqrt{m_i}} \left(e_i e_i^\top B M^{1/2} + M^{1/2} B e_i e_i^\top \right).$$

For a simple eigenvalue with unit eigenvector v , first-order perturbation gives $d\lambda_{\max}/dm_i = v^\top (dA/dm_i)v$. The two terms are transposes of one another and A is symmetric, so they contribute equally, giving $d\lambda_{\max}/dm_i = (BM^{1/2}v)_i v_i / \sqrt{m_i}$. At the symmetric point $M = m_1 I$ we have $A = m_1 B$, so v is the leading eigenvector of B and $BM^{1/2}v = \sqrt{m_1} \lambda_{\max}(B)v = \sqrt{m_1} N_{\text{eff}} v$, whence $d\lambda_{\max}/dm_i = N_{\text{eff}} v_i^2$. Summing over i gives N_{eff} , because v is a unit vector. Under (W3) an exchangeable B commutes with every permutation matrix, so a simple leading eigenvector must be permutation invariant, giving $v_i^2 = 1/N$ and $dm_N/dm_i = N_{\text{eff}}/N$. $\square$

Certificate C34 checks this by finite differences on random alignment matrices, confirming that the shares sum to N_{eff} and equal N_{eff}/N on exchangeable R . It is the load-bearing certificate of this section: everything else in the theorem is arithmetic on top of the marginal crowding share, and until C34 that share was the step where a plausible-looking $1/N$ could have been wrong by a factor of N_{eff} .

Two things are worth reading off. The *sum* of the marginal effects is N_{eff} and not 1, so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count. And an *individual* firm moves m_N by N_{eff}/N , so as N grows each firm's own footprint shrinks while the total grows, which is the arithmetic shape of every commons problem.

F.3 The two problems and the wedge

Firm i bears weight w_i on common-mode variance and the planner bears W :

$$\text{private: } \max_{a_i} B(a_i) - w_i V(m_N(a)), \quad \text{social: } \max_a \sum_j B(a_j) - W V(m_N(a)).$$

Write $V'(m) = 2\sigma^2 m / (1 - m^2)^2 > 0$ on $(0, 1)$ and evaluate at a symmetric profile $a_i = a$, so that $m_N = \mu(a) N_{\text{eff}}$. Chaining (W1) with Lemma 7 gives $dm_N/da_i = (N_{\text{eff}}/N) \mu'(a)$ at the symmetric profile, and the two first-order conditions are

$$(P) \quad B'(a) = w_i V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad (S) \quad B'(a) = W V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a).$$

They are the same equation with a different price on crowding, and the theorem is the gap between the two coefficients.

Proof of Theorem 8. Subtract (P) from (S). A firm facing a per-unit fee t on aggressiveness solves $\max B(a_i) - w_i V(m_N(a)) - ta_i$, whose first-order condition is (P) with $B'(a)$ replaced by $B'(a) - t$. Substituting

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a) = (W - w_i) \frac{2\sigma^2 m_N}{(1 - m_N^2)^2} \cdot \frac{N_{\text{eff}}}{N} \mu'(a)$$

reproduces (S) term for term, so the taxed decentralized equilibrium implements the symmetric social optimum. $\square$

With symmetric weights $w_i = w$ and $W = \chi N w$ for the $\chi > 1$ that (W4) supplies, $t^* = (\chi N - 1) w V'(m_N) (N_{\text{eff}}/N) \mu'(a)$ against a total marginal cost of $\chi N w V'(m_N) (N_{\text{eff}}/N) \mu'(a)$, so the ignored fraction is $(\chi N - 1)/(\chi N)$. Setting $\chi = 1$, which shuts off client exposure entirely and leaves only the firm-to-firm channel, the ignored fraction is exactly $(N - 1)/N$. Note that N_{eff} cancels from this ratio, so the $(N - 1)/N$ reading is a generic commons result and it is the provenance asymmetry below that is specific to this paper.

Proposition F.1. t^* is strictly increasing in N , in κ and in m_N , and diverges as $m_N \rightarrow 1$.

Proof. Take the supply-chain specialization $N_{\text{eff}} = 1 + \kappa s(N - 1)$ with symmetric weights, so that $t^* = (\chi - 1/N) w V'(m_N) N_{\text{eff}} \mu'(a)$. The factor $\chi - 1/N$ is strictly increasing in N ; N_{eff} is strictly increasing in N whenever $\kappa s > 0$ and in κ whenever $s > 0$ and $N \geq 2$; $m_N = \mu(a) N_{\text{eff}}$ inherits both; and $V'(m) = 2\sigma^2 m / (1 - m^2)^2$ is strictly increasing on $(0, 1)$ and diverges at 1, since both m and $(1 - m^2)^{-2}$ are. Every factor moves the same way, so the product does. $\square$

872 The divergence is the economically loaded part. The fee that would correct a market is not proportional
873 to how crowded it is; it blows up like $(1 - m_N)^{-2}$ as the market approaches its stability boundary, so
874 a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too
875 low by the time the market is close to the edge. Certified in C20.

876 *Proof of Corollary 9.* Write $C(a) = V(m_N(a))$ and

$$F_{\text{priv}}(a) = B'(a) - wC'(a), \quad F_{\text{soc}}(a) = B'(a) - WC'(a).$$

877 Both are strictly decreasing, since B' is strictly decreasing by (W2) and C' is nondecreasing by
878 convexity, so each has at most one zero and by the interior assumption exactly one, namely a_d and a_s .
879 Because $C'(a) > 0$ on the stable range, as $V' > 0$, $N_{\text{eff}} > 0$ and $\mu' > 0$, and because $W > w$ by
880 (W4),

$$F_{\text{soc}}(a) = F_{\text{priv}}(a) - (W - w)C'(a) < F_{\text{priv}}(a) \quad \text{for every } a.$$

881 Evaluating at a_d gives $F_{\text{soc}}(a_d) < F_{\text{priv}}(a_d) = 0$, and since F_{soc} is strictly decreasing and vanishes
882 at a_s , this forces $a_s < a_d$. Setting $\chi = 1$ so that $W = Nw$ gives $W - w = (N - 1)w > 0$ exactly
883 when $N \geq 2$, so the strict inequality survives with no client-side channel at all. At $N = 1$ with $\chi = 1$
884 the wedge is zero and the two problems coincide, which is the correct degenerate case: a single firm
885 bearing its own variance in full internalizes everything. $\square$

886 The distortion grows as the market crowds, since the coefficient gap $(W - w)C'(a)$ is proportional
887 to $V'(m_N)N_{\text{eff}}$, both of which increase in N and in κ . Certified in C21, including the vanishing gap
888 at $N = 1$.

889 **F.4 The provenance channel**

890 **Proposition F.2.** Under the supply-chain specialization, $dm_N/ds = m_1\kappa(N - 1)$ exactly, and the
891 corresponding adoption wedge is $t_s^* = (W - w_i)V'(m_N)m_1\kappa(N - 1)$.

892 *Proof.* $m_N = m_1(1 + \kappa s(N - 1))$ is affine in s , so the derivative is the coefficient, and the wedge
893 follows from the proof of Theorem 8 with s in place of a_i as the choice variable. $\square$

894 The same wedge therefore prices shared-model adoption directly, which turns the monoculture
895 externality into a fee rather than a warning. The contrast with the aggressiveness channel is the
896 paper-specific result. The aggressiveness derivative carries a factor N_{eff}/N , bounded above by
897 $(1 + \kappa(N - 1))/N$ and therefore tending to κ from above as N grows, so a firm's marginal footprint
898 through its own cadence does not grow without bound. The provenance channel is linear in N and
899 does not decay. A firm's decision to adopt the market-leading vendor imposes a marginal cost on
900 the market that grows with market size while its private quality gain does not, and that asymmetry is
901 why unregulated adoption dynamics run toward the unstable configuration rather than away from it,
902 and why a diversity floor is a different instrument from a cadence cap rather than a restatement of it.
903 Certificate W6 measures the provenance channel growing from 0.79 to 87.91 between $N = 10$ and
904 $N = 1000$ while the aggressiveness channel falls from 0.82 toward $\kappa = 0.8$.

905 The three levers are the standard instrument taxonomy: cadence caps are quantity regulation, correc-
906 tion mandates are a technology mandate, diversity floors are a structural remedy, and the wedge is a
907 price instrument. The choice between prices and quantities under uncertainty is Weitzman's question,
908 which this paper cites for framing rather than solves.

909 **G Supervision from public prices**

910 Every quantity in Theorems 2 through 8 involves R and s , which no outsider observes. Supervisors
911 do not inspect firms' models, training sets or vendor contracts, so a theory whose every variable is
912 unobservable to the regulator it addresses is a theory of a problem rather than a theory a regulator can
913 act on.

914 Near the boundary the system tells on itself. As $m_N \rightarrow 1$ the aligned combination of firms' decisions
915 exhibits critical slowing down: the lag-1 autocorrelation of the leading principal component of public
916 quotes approaches one in magnitude, negative because the common mode oscillates, and its variance

917 share grows like $1/(1 - m_N)$. A supervisor can therefore estimate distance to instability from public
 918 prices alone, with no access to any firm’s model, data or code, and the estimator sharpens exactly
 919 as the market approaches the boundary, which is the opposite of the usual situation in which risk
 920 measurement degrades in the regime that matters.

921 This material is deliberately thin and the body flags it as deferred. The body carries the estimator
 922 and the consistency claim in one paragraph. The empirical panel, namely the leading component’s
 923 variance share and lag-1 autocorrelation of cross-sectional spread co-movement overlaid on the
 924 model-implied fragility index, together with a placebo on series where no dealer-model channel
 925 exists, is scoped and is not run for this submission. It would be framed as consistency evidence
 926 and never as identification, since spread co-movement is contaminated by common macro shocks
 927 and the data are public proxies rather than trade-level records. The consistency proof, the central
 928 limit theorem and the detection sample complexity are journal material. Writing the estimator down
 929 formally, rather than as the description above, is the first open item.

930 H Certificates

931 Every closed form in the body carries a numerical certificate. A certificate is an assertion-based script
 932 that re-derives the expectation independently, by dense eigensolve, finite difference, brute force or
 933 direct simulation, rather than importing the theory module’s own answer, and that exits nonzero on
 934 any disagreement. A closed form reaching the paper without a passing certificate ships as derived
 935 and never as verified.

	Assertions	File	What it establishes
	123	<code>verify_theorem1_proof.py</code>	the reduction, the properties of R , the spectrum, the mean bound, the heterogeneous-modulus bound, and the supply-chain concentration rate
	60	<code>verify_theorem1_anchors.py</code>	the three anchors, the clustered topology, and the signed error of the mean index
	59	<code>verify_theorem2_cadence.py</code>	the inner contraction, hypothesis (C), the joint map, the frontier in both forms, and the realized dynamics
936	70	<code>verify_theorem3_herd_immunity.py</code>	the two-block quadratic, the degenerate blocks, monotonicity in θ , the imperfect-correction law, the critical efficacy and the integer threshold
	125	<code>verify_theorem4_wedge.py</code>	the AR(1) reduction and its sign convention, Lemma 7 by finite differences, the comparative statics, and over-adaptation
	56	<code>verify_theory_module.py</code>	acceptance tests on the closed-form module
	32	<code>verify_hetero_env.py</code>	the reference environment, including its reduction to the homogeneous market at $R = \mathbf{11}^\top$
	525	total	all passing at submission

937 Two of these were falsifying rather than confirming. C18 showed that the strong-correction limit
 938 errs optimistic, which changed what Theorem 5 claims and moved the exact two-block root from an
 939 optional refinement to the theorem itself. C14 showed that the clamped restatement of the blind-firm
 940 threshold is not equivalent to the primitive form, failing on 134 of 4000 draws at the all-blind corner.
 941 Both corrections are recorded above, where they apply.

942 I Experimental specifications

943 **Environments.** There are two, and the distinction between them is what the status flags track.
 944 The *order-flow simulator* is a genuine N -dealer market with informed flow, spread and inventory,
 945 which reduces bit for bit to the single-dealer market at $N = 1$. The *reference environment* places
 946 firms at chosen response directions and lets them retrain against a shared pool, realizing the response
 947 geometry of Lemma 1 and nothing else: no informed flow, no spread, no inventory, no microstructure
 948 of any kind. Its acceptance test, run before any measurement is taken from it, is that at $R = \mathbf{11}^\top$ it
 949 reproduces the homogeneous market on three counts, namely a measured alignment of exactly $\mathbf{11}^\top$,
 950 an N_{eff} of exactly $1 + \kappa(N - 1)$, and a built Jacobian equal to (1) to 10^{-10} .

Protocol. Inherited from the base model and non-negotiable, since violating any of these silently invalidates a panel: sweep the feedback gain and never the confounded adversariality parameter; probe at the operating spread with common random numbers; scale the liquidity boost per the environment’s guidance when multi-dealer runs approach the informed-flow cap, and never de-saturate silently; and treat beyond-boundary probe readings as diagnostics rather than as slopes. All runs are CPU-only and deterministic from a (config, seed) pair. Stability is estimated from the trajectory’s asymptotic growth rate under the actual retraining map rather than read off an eigensolve, so a panel can disagree with its closed form.

The panels. Panel 1 sweeps N in the homogeneous shared-pool market and reproduces the amplification reported in a recent preprint. It is the paper’s one measured result, and what it validates is the monoculture corner that Theorem 2 generalizes rather than any of the four results built on it. Panel 2 measures m_N over a grid of firms by shared-model fraction against the predicted boundary, with firms placed at the target alignment exactly rather than in distribution so that an s sweep moves along the exact curve; its clustered companion, three aligned firms among ten, is the topology worked in Appendix B. Panel 3 sweeps the (N_{eff}, K) grid against K_{max} at $c = 0.8$, including the critical-crowding column where the window closes. Panel 4 runs mixed markets at corrected fractions $0, 1/N, \dots, 1$ in an unstable market. Panel 5 traces the (ρ, s) iso-stability curve, and it is the paper’s most policy-legible object. Panel 6 solves both first-order conditions by bisection with m_N read from the reference environment’s realized dynamics rather than from the closed form, so the over-adaptation verdict is a property of the simulated market.

Why five of six are dry runs. A heterogeneous-response port of the order-flow simulator was built and is exact at flat exposure profiles, reproducing the unmodified simulator at every N from 1 to 6 at relative error 0.00. Its acceptance gate then failed. The liquidity and price-impact channels stay shared whatever the response profiles are, leaving a residual reaching 17.7 percentage points across the separation range at $N = 2$, larger than several effects these panels resolve, and at $N \geq 4$ the finite-difference probe no longer measures a local slope. Two repairs are identified, routing liquidity and impact through the profiles as well, or finding a configuration whose probe stays local at the N the panels need, and neither is attempted for this submission, since both are multi-day edits and the first risks the bit-for-bit anchor that panel 1 rests on. No figure is drawn from that port and no status was moved. Panel 6 is a dry run for a different reason: the order-flow simulator carries no aggressiveness choice variable and no welfare object, so a measured counterpart does not exist to be built.

J Deferred items

Stated so that the boundary of what is proved is legible. The fully heterogeneous case without (H1), which is the Kronecker object of Appendix B, and share-weighted alignment for unequal-sized firms, which needs a majorization condition on the share vector since the naive claim that any reweighting raises λ_{max} is false. Non-separable spillover κ_{ij} , which folds into R only when it factors. The sharp operator-norm rate $\sqrt{N}/d$ of Appendix C, which needs a matrix-Bernstein argument rather than the Frobenius step, and a version of design assumption (D) under dependence across the idiosyncratic components, which is what shared data rather than shared models would produce. Asynchronous and heterogeneous cadences, which break the eigenvector sharing that Theorem 4 uses. Three or more correction levels, which break the two-block collapse and return a genuine secular equation. A corrected firm that also changes its response direction, which would move R and which hypothesis (M) does not model. And beyond the symmetric point, the observation that $dm_N/dm_i = N_{\text{eff}}v_i^2$ makes the marginal crowding share the leading eigenvector’s participation ratio at firm i , so a firm sitting on the crowded mode imposes more than its $1/N$ share, and the asymmetric wedge that follows is a genuine open question.

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Justification: The public repository names an author, so it is withheld at submission to preserve the blind. The derivations are in Appendices A–F and the certificate inventory in Appendix H, both in this PDF. The seven certificate files themselves and the panel code go to reviewers as anonymized supplementary material, and the repository is linked in the camera-ready version; Section 8 states this arrangement. No dataset is used: every panel is simulated and deterministic from a (config, seed) pair.

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